Multiplier-Free Seizure Detection with a LUT-Native Neural Network on Behind-the-Ear EEG, ECG, EMG and Motion

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Abstract*—Wearable seizure detection has to run for days on a small battery, which favours classifiers without multipliers. We study LUT-native neural networks, whose neurons are FPGA lookup tables, for patient-independent seizure detection from two-channel behind-the-ear EEG, ECG, EMG and motion in the SeizeIT2 dataset (125 patients, 11,626 hours, 883 seizures). A bit-exact integer front end computes 117 features without multipliers; a thermometer encoder and a Differentiable Weightless Neural Network (DWN) of 50 six-input LUTs turn them into a popcount score; synthesised for the Zynq UltraScale+ fabric, the whole classifier takes 405 LUTs and 454 flip-flops, with no DSP or block RAM. On five-fold cross-validation over the 96 training patients the DWN reaches an event-F1 of 0.211 with 2.4 false alarms per day at about 960 estimated LUTs and no multipliers, seven to eight times fewer LUTs than TreeLUT and a logic-gate network, and better than an INT8 network that needs 9,600 multiplications per decision, though below gradient-boosted trees (0.292). Training with random sensor dropout keeps the detector working when the ECG is missing (event-F1 0.172 against 0.092, $p = 0.005$), and distillation from LightGBM cuts false alarms from 2.4 to 0.9 per day ($p = 3 \times 10^{-7}$). On the 29 held-out test patients, scored once with 5 seeds, the final DWN reaches an event-F1 of 0.248 ± 0.016 with 0.8 false alarms per day at about 900 LUTs, against 0.027 and 106 false alarms per day for the clean ChronoNet baseline ($p = 0.027$), 0.233 for TreeLUT (6.3k LUTs) and 0.308 for LightGBM. With the dataset paper's metrics, a pre-registered two-stage variant that confirms each alarm with a second 50-LUT network detects 73.3 % of seizures at 10.9 false alarms per hour, against 71.1 % at 11.0 for the published SVM. The design runs on a Kria KV260: the integer front end in C on the Arm cores reproduces all 117 features bit for bit, and the classifier in the fabric reproduces the software on all 431,960 test windows in simulation. We also report an independent reproduction of the published ChronoNet baseline, whose released code selects its model on the test patients.

I. Introduction

Patients are often unaware of their own seizures, so diaries undercount them [1], and wearable detection is clinically recommended today only for tonic-clonic seizures [2]. Focal seizures that do not become tonic-clonic (822 of the 883 in SeizeIT2) remain the open problem, with false alarms the main obstacle [3]. Behind-the-ear (bte) EEG is discreet but sees only two channels; adding ECG, EMG and motion helps [4], [5], but the detectors that fuse them are floating-point software or multiply-accumulate networks on microcontrollers [6].

LUT-native networks replace weighted sums by lookup tables: each neuron is a Boolean function of a few input bits that maps one-to-one onto an FPGA LUT. DWNs [7], logic-gate networks [8] and LUT networks [9] train such networks by gradient descent, and tree ensembles can be mapped to LUTs in the same spirit [10]. They have been applied to ECG arrhythmia [11] and EEG classification [12], but not to seizure detection, which brings extreme class imbalance (0.12 % seizure time), patient-to-patient variability, missing sensors and event-level clinical scoring.

Contributions. (i) The first differentiable LUT-native seizure detector and the first benchmark of LUT-native models (DWN, logic-gate network, TreeLUT) against standard machine learning on the official SeizeIT2 split with SzCORE event scoring [13]. (ii) A bit-exact, multiplier-free integer front end for bte-EEG, ECG, EMG and motion, with a fixed-point specification and a reference Verilog classifier that matches the software on all test vectors. (iii) Distillation from a tree ensemble into a LUT network, which cuts false alarms, and modality dropout, which makes it robust to a missing ECG, selected among seven families of extensions by per-patient tests. (iv) An independent reproduction of the SeizeIT2 ChronoNet baseline, with a protocol that uses no test information. (v) A verified implementation on a Kria KV260 with the synthesised cost of all three LUT-native models under one flow.

II. Related Work

Wearable bte-EEG. Two bte channels capture many focal seizures [14]; ECG raises sensitivity by 8–11 % [4]. Tree ensembles won the ICASSP 2023 wearable-EEG challenge [15], [16]. The SeizeIT2 paper reports SVM and ChronoNet [17] baselines [18]. *Hardware detectors.* ASICs reach nJ–µJ per classification with SVMs and trees [19]–[21]; an SNN on a small iCE40 FPGA detects seizures on four scalp channels [22]; transformers and fusion networks run on microcontrollers [6], [23]. Almost all use EEG only, patient-specific training and epoch-level metrics. *LUT-native networks.* Beyond the works above, the thermometer encoder can dominate the cost of small DWNs [24], and a recurrent logic-gate network exists for language [25]. The only weightless seizure work is a WiSARD study on four patients [26].

III. Dataset and Protocol

Data. SeizeIT2 (OpenNeuro ds005873 v1.1.0) [18]: 125 patients with focal epilepsy from five European monitoring units, 2,850 recordings, 11,626 hours of two-channel bte-EEG, single-lead ECG and EMG at 256 Hz and accelerometer and gyroscope at 25 Hz; ECG/EMG are missing in 3.1 % and

motion in 3.9 % of the time. Our audit found 883 seizures (702 in the 96 training patients, 181 in the 29 test patients; the paper states 886), and 256 Hz rather than the documented 250 Hz. Seizures were annotated on video and full-scalp EEG. *Split and tuning.* Official split: training sub-001–096, test sub-097–125. Every choice (features, hyperparameters, thresholds, post-processing, extensions) is made by five-fold patient-wise cross-validation inside the training patients; each final model is then trained on all 96 and scored once per seed on the test patients. *Scoring.* SzCORE event scoring [13] (any overlap, 30 s pre- and 60 s post-ictal tolerance, events closer than 90 s merged), averaged over patients, with false alarms per 24 h. A decision is made every second; an alarm is raised when at least k of the last 10 decisions are positive; the threshold, k and an artefact gate are chosen on pooled out-of-fold predictions. Per-patient paired Wilcoxon tests and patient-bootstrap 95 % intervals; an extension is kept only if $p < 0.05$ in its favour.

IV. Method

A. Integer multimodal front end

Samples arrive as int16 (0.25 µV per count). Every filter is a second-order section with coefficients rounded to at most five signed powers of two and second-order error feedback; without it the rounding error of the 0.5 Hz high-pass exceeds the EEG itself. Every second, 2-s windows give per-channel and cross-channel EEG magnitude, line length, mobility, zero crossings, extrema and four band magnitudes; ECG heart rate and RR variability in the log domain (no division); EMG magnitude, line length, zero crossings and a band ratio; accelerometer and gyroscope magnitude, jerk, activity and band magnitude; per-modality presence bits; and, for most features, the difference from a causal 64-s moving average. On 10 training patients the integer features match a floating-point twin (median correlation 0.9965; LightGBM AUROC 0.736 against 0.735).

B. Thermometer encoding and DWN

Each of the 40 selected features is compared against 8 thresholds at training quantiles, $b_{f,i} = [x_f > t_{f,i}]$. A layer of 50 LUT6 neurons reads these bits through a learnable mapping and is trained with extended finite differences [7]; the score is the popcount of the seizure group minus that of the background group. Features are ranked inside each training set by the gain of a LightGBM model; this replaced mutual information after an interpretability study showed that mutual information ignored the heart-rate features (event-F1 0.148 $\rightarrow$ 0.208, $p = 0.047$). Estimated cost: about 850 LUTs of comparators, 50 network LUTs, 57 popcount and 12 post-processing LUTs.

C. Extensions and modality dropout

We tested seven families of extensions against this base: a stateful network with 16 state bits fed back each second and stacked windows (N1), encoder designs (N2), label-free per-patient threshold adaptation (N3), distillation from LightGBM and teacher-guided hard negatives (N4), focal, noise-robust and multiple-instance losses and onset weighting (N5), modality dropout, reliability-gated and late fusion (N6), and alarm interpretation (N7). Two were adopted. *Distillation*: a LightGBM teacher trained on the same windows provides soft targets (weight 0.5, temperature 2). *Modality dropout*: during training, ECG, EMG and motion are each removed independently with probability 0.2 per window, exactly as when the sensor is absent (features and presence bits set to zero).

TABLE I
FIVE-FOLD CROSS-VALIDATION ON THE 96 TRAINING PATIENTS (SZCORE, MEAN OVER PATIENTS; BEST-F1 OPERATING POINT; HARDWARE COST ESTIMATED).

Model	AUROC	F1	Sens	FA/24h	Cost
LightGBM	0.870	0.292	0.327	3.6	≈30.6k LUTs
CatBoost	0.863	0.286	0.337	3.0	≈27.7k LUTs
XGBoost	0.867	0.277	0.286	3.4	≈47.9k LUTs
TreeLUT (5-bit leaves)	0.857	0.266	0.329	4.2	≈6.4k LUTs
DWN, 50 LUT6	0.840	0.211	0.315	2.4	≈960 LUTs
DWN + distillation (final)	0.851	0.229	0.267	0.9	≈870 LUTs
DWN + modality dropout	0.849	0.188	0.275	1.9	≈945 LUTs
Random forest	0.852	0.198	0.216	0.5	≈323k LUTs
Logic-gate network	0.843	0.171	0.330	4.9	≈7.5k LUTs
INT8 MLP	0.830	0.147	0.225	7.8	9,600 mult.
Logistic regression	0.806	0.105	0.216	8.8	117 mult.

V. Results

A. Baseline reproduction

On the 29 test patients, our PyTorch port of ChronoNet with the released protocol detects 74.5 % of seizures at 22.6 false alarms per hour (published: 84.2 % at 100.5) with AUROC 0.741 (0.792); the per-recording AUPR (0.109) matches the published 0.104. The released code selects the training epoch on the test patients; a clean run (last epoch) gives 67.1 % at 11.9 false alarms per hour, and under SzCORE an event-F1 of 0.027 at 106 false alarms per day. The clean run is our ChronoNet reference.

B. Model comparison (training patients)

Table I compares the model families on cross-validation. The DWN is the most compact model, with one of the lowest false-alarm rates, and beats the INT8 MLP on every metric; it remains below gradient-boosted trees (per-patient event-F1 against LightGBM, $p = 0.004$). Between 50 and 200 LUTs, size did not change event-F1 significantly; four thresholds per feature were worst at every size.

C. Extensions and missing sensors

None of the N1, N3 and N5 extensions improved event-F1 (Table II); the stateful network and onset weighting cut false alarms ($p = 0.013$, $p = 0.014$) at lower sensitivity, and the focal and noise-robust losses were significantly worse. The base loses most of its accuracy without ECG or motion (Table III). Modality dropout keeps it working: without ECG its event-F1 is 0.172 against 0.092 for the base ($p = 0.005$), without motion 0.192 against 0.136 ($p = 0.021$), at no significant cost

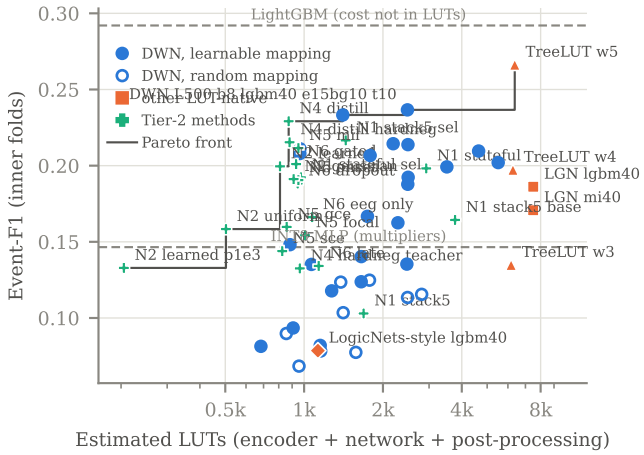


Fig. 1. Event-F1 against estimated LUTs on the training folds, with the Pareto front.

TABLE II
TIER-2 METHODS AGAINST THEIR OWN BASELINES (INNER FOLDS; MEAN PER-PATIENT DIFFERENCE, BOOTSTRAP 95% INTERVAL, PAIRED WILCOXON).

Method	ΔF1 [95% CI]	p	ΔFP/24h	p	ΔLUTs
n1_stack5_sel	+0.006 [-0.041, +0.049]	0.25	-0.3	0.81	+478
n1_stateful_sel	-0.019 [-0.066, +0.026]	0.66	-1.3	0.013	+8
n2_gaussian	-0.020 [-0.057, +0.017]	0.25	-0.3	0.0047	-52
n2_learned	-0.012 [-0.054, +0.033]	0.52	-1.4	1.1e-08	-155
n2_learned_p1e3	-0.078 [-0.128, -0.029]	0.0013	+4.7	3.6e-12	-759
n2_uniform	-0.053 [-0.105, -0.000]	0.038	-0.3	1.6e-08	-461
n46_distill_dropout	-0.016 [-0.055, +0.024]	0.2	-1.0	0.0062	-113
n4_distill	+0.018 [-0.031, +0.067]	0.26	-1.5	3.3e-07	-91
n4_distill_hardneg	+0.004 [-0.038, +0.049]	0.7	+0.4	0.47	-84
n4_hardneg_teacher	-0.079 [-0.126, -0.031]	0.00056	+1.7	0.0052	-1
n5_focal	-0.057 [-0.106, -0.012]	0.038	-0.9	0.028	+44
n5_gce	-0.051 [-0.086, -0.019]	0.0052	-0.8	0.27	-106
n5_mil	+0.000 [-0.038, +0.037]	0.91	-0.6	0.11	-13
n5_onset	-0.021 [-0.058, +0.016]	0.28	-1.1	0.014	-13
n5_sce	-0.067 [-0.111, -0.027]	0.0031	-0.8	0.6	-139
n6_dropout	-0.023 [-0.066, +0.019]	0.23	-0.5	0.44	-19
n6_eeg_only	-0.045 [-0.096, +0.003]	0.084	+1.2	0.0014	+109
n6_gated	-0.010 [-0.062, +0.038]	0.92	-0.9	0.11	-32
n6_late	-0.077 [-0.125, -0.027]	0.00063	-1.9	4.1e-09	+173

when every sensor is present ($p = 0.23$). Late fusion, one small DWN per modality, collapsed whenever a modality was missing. Distillation from LightGBM cut false alarms from 2.4 to 0.9 per day ($p = 3 \times 10^{-7}$) with a slightly higher event-F1 (0.229, $p = 0.26$) and became the final model; teacher-guided hard negatives and the encoder variants (uniform, Gaussian and learned thresholds) were not better. Combining distillation with modality dropout was significantly worse with all sensors present than distillation alone ($p = 0.030$), so the two are evaluated as separate models.

D. Seizure types

Out of fold, the base detects 29 of 33 focal to bilateral tonic-clonic seizures (88 %), but 17 % of focal impaired awareness, 9 % of focal aware and 8 % of focal seizures with unknown awareness.

TABLE III
EVENT-F1 WITH SENSORS REMOVED AT TEST TIME (TRAINING FOLDS, EACH MODEL AT ITS OWN OPERATING POINT).

Model	intact	no ECG	no EMG	no motion	EEG only
DWN base	0.211	0.092	0.199	0.136	0.074
+ modality dropout	0.188	0.172	0.190	0.192	0.112
late fusion	0.134	0.070	0.012	0.056	0.000

TABLE IV
HELD-OUT TEST PATIENTS (29): MEAN ± STD OVER 5 SEEDS, SZCORE EVENT METRICS AT THE BEST-F1 OPERATING POINT CHOSEN ON THE TRAINING FOLDS; p: PAIRED WILCOXON ON PER-PATIENT EVENT-F1 AGAINST CLEAN CHRONONET.

Model	F1	Sens	FA/24h	p vs ChronoNet	Cost
DWN + distillation (final)	0.248±0.016	0.303	0.8	0.03	895 LUTs
DWN base	0.172±0.044	0.340	2.7	0.01	980 LUTs
DWN + modality dropout	0.242±0.030	0.272	0.4	0.03	951 LUTs
LightGBM	0.308±0.013	0.331	0.6	0.03	30,750 LUTs
XGBoost	0.294±0.014	0.301	0.4	0.06	48,295 LUTs
CatBoost	0.254±0.009	0.330	1.2	0.03	27,720 LUTs
Random forest	0.246±0.003	0.245	0.2	0.3	346,576 LUTs
Logistic regression	0.123±0.002	0.255	5.1	0.6	117 mult.
TreeLUT (5-bit leaves)	0.233±0.044	0.292	1.0	0.02	6,330 LUTs
INT8 MLP	0.155±0.008	0.249	1.7	0.02	9,600 mult.
Logic-gate network	0.191±0.021	0.291	1.4	0.03	7,452 LUTs
ChronoNet (clean)	0.027	0.790	106.3	–	CNN+GRU

E. Held-out test patients

The design was frozen before any of our models was scored on a test patient. Table IV gives the result. The final DWN reaches an event-F1 of 0.248 ± 0.016 with 0.8 false alarms per day, above its cross-validation value (0.229): the improvement from distillation holds on unseen patients (base DWN 0.172). It is above TreeLUT (0.233) at a seventh of the LUTs, above the logic-gate network (0.191) and the INT8 MLP (0.155), level with random forest (0.246) and CatBoost (0.254), and below LightGBM (0.308, $p = 0.039$ per patient). Against clean ChronoNet (event-F1 0.027, 106 false alarms per day) it is better per patient ($p = 0.027$): ChronoNet detects more seizures (79 %) but raises over a hundred times more false alarms. With the dataset paper's own metrics at a matched false-alarm rate, the final DWN detects 43.2 % of seizures at 1.24 false alarms per hour, against 34.8 % for ChronoNet at 1.26 per hour (a threshold read from its own test curve, which favours it), and its pooled window AUROC is 0.883 against 0.746. The modality-dropout variant reaches 0.242 with 0.4 false alarms per day.

F. Comparison with the dataset paper

Fig. 2 places the frozen models on the dataset paper's sensitivity–false-alarm plane. Between 0.02 and 10 false alarms per hour the final DWN detects more seizures than ChronoNet at every rate and stays level with LightGBM and TreeLUT; at the SVM's 11 false alarms per hour its curve reaches 69.7 % (published SVM: 71.1 %; read from the test curve, hence optimistic). As a secondary analysis decided after the final test, we added a post-ictal verification stage: the DWN runs at a sensitive threshold, and 96 s after each candidate alarm a second 50-LUT DWN compares 16-s ictal and 64-s post-ictal windows of 45 features with the preceding 128 s

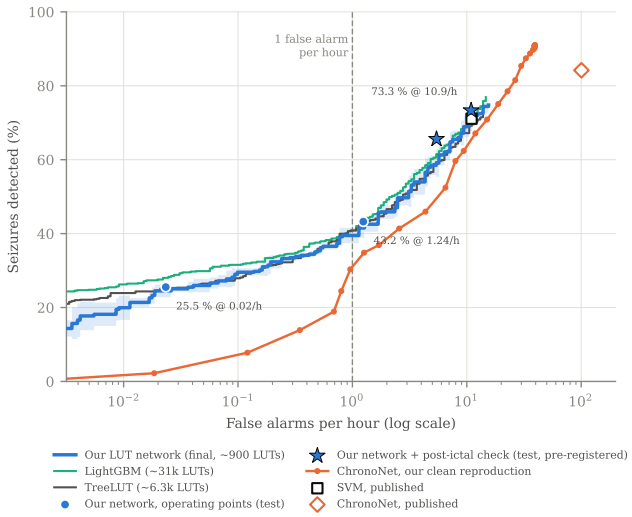


Fig. 2. Seizures detected against false alarms per hour on the 29 test patients (the dataset paper's metrics). Curves: threshold sweeps of the frozen models (mean of 5 seeds); dots: operating points chosen on the training folds; stars: two-stage detector with post-ictal verification (pre-registered); open markers: published values.

and confirms or rejects it. All its settings were chosen on the training folds (81.9 % at 9.6 false alarms per hour out of fold) and committed before a single test evaluation, which gave 73.3 ± 1.4 % at 10.9 ± 0.6 false alarms per hour over five seeds (all seeds above 71.1 %; two of five above 11.0 false alarms per hour). Verification adds 3.6–6.5 points over the DWN alone at the same rate; it suits seizure logging rather than instant alarms, and like the paper's baselines this is a high-false-alarm regime (event-F1 0.027).

VI. Hardware

Partitioning. On a Kria KV260 (Zynq UltraScale+ XCK26), the integer front end runs in C on the Arm Cortex-A53 cores and the classifier runs in the fabric behind an AXI4-Lite interface at 100 MHz (Fig. 3): feature registers, a commit register, the frozen operating point and a second, programmable one, counters, and a built-in self-test that streams 1,024 golden vectors from block RAM at one window per clock and checks every output in hardware. *Verification.* The C front end reproduces all 117 integer features and both artefact flags of all 431,960 windows of the 10 exported test recordings exactly. The classifier, driven through its AXI4-Lite registers exactly as the board software drives it, reproduces score, decision and alarm on all 431,960 windows at three operating points (Verilator; deliberately corrupted vectors are caught); latency is 3 cycles and the self-test runs 1,024 windows in 1,028 cycles. *Cost.* Table V gives synthesis results for the three LUT-native models, each generated from its frozen final model, checked bit-exact against it in simulation, and synthesised by the same flow. The DWN needs 405 LUTs, 6.7 times fewer than TreeLUT and 14.2 times fewer than the logic-gate network; the analytical estimates used in Table I were conservative for all three (895, 6,330 and 7,452 LUTs), so the ranking holds. [Phase 2: Vivado post-implementation utilisation, maximum frequency and power of the three models and of the KV260 design; on-board bit-exact replay, latency and the SOM power measured with the INA260 monitor; front-end time per hour of signal on the Cortex-A53.]

TABLE V
Synthesised cost of the LUT-native classifiers (Zynq UltraScale+; Yosys `synth_xilinx`, the same script for all three; each includes input registers, comparators, model logic, score, decision and the 10-s alarm).

Model	LUTs	FFs	DSP	Event-F1 (test)
DWN + distillation (ours)	**405**	454	0	0.248
TreeLUT (5-bit leaves)	2,721	1,455	0	0.233
Logic-gate network	5,758	2,200	0	0.191

Sensors → int16 samples → **Arm A53**: integer front end (C, 117 features) → AXI4-Lite → **fabric**: 146 comparators → 50 LUT6 → popcount → score, decision, k-of-10 alarm (+ programmable operating point, BIST)

Fig. 3. SeizeLUT on the Kria KV260.

VII. Discussion and Limitations

Accuracy per LUT favours the DWN by a wide margin, and the thermometer encoder, not the network, dominates its cost, as also measured by [24]. Focal seizures remain mostly undetected, consistent with the clinical evidence that wearable detection is established only for tonic-clonic seizures [2]. Results come from retrospective recordings of one dataset, and the public release has no centre or age labels, so no centre or age breakdown is possible. The post-ictal analysis was added after the final test; it was pre-registered on the training folds, but it is a second look at the same test patients.

VIII. Conclusion

A 50-LUT weightless network on integer multimodal features detects seizures across patients with no multipliers in 405 synthesised LUTs. On 29 held-out patients it reaches an event-F1 of 0.248 with 0.8 false alarms per day, above a published deep-learning baseline and TreeLUT and level with random forests; distillation and modality dropout address false alarms and missing sensors.

Reproducibility

Code, configurations, decision log, export packages and the Verilog generator are released with the paper. Dataset: SeizeIT2, DOI 10.18112/openneuro.ds005873.v1.1.0.

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